



RAMCO INSTITUTE OF TECHNOLOGY

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NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Artificial Intelligence and Data Science

Academic Year 2025 – 2026 (Odd / Even Semester)

Degree, Semester & Branch: BTech/V/AI&DS B Section

Course Code & Title: AD3501-Deep Learning

Name of the Faculty member (s): Dr R M Rajeshwari

Innovative Practice Description

- **Unit / Topic:** Unit IV / Grid Search
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 35
- **Activity Chosen:** Reciprocal Teaching

Date: 09/10/2025

- **Justification:**

Grid Search was chosen in the reciprocal teaching methodology because it helps students actively explore how hyperparameter tuning improves model performance through guided discussion and experimentation. By systematically testing parameter combinations, learners can observe cause-effect relationships, fostering deeper conceptual understanding. This collaborative approach enhances problem-solving and critical thinking in machine learning model optimization.

- **Time Allotted for the Activity:** 45 Minutes
- **Details of the Implementation:**

The implementation of Grid Search using the reciprocal teaching methodology involves engaging students actively through four key stages—prediction, clarification, questioning, and summarization—combined with hands-on experimentation. Initially, students are asked to predict which hyperparameters, such as learning rate, kernel type, or the number of estimators, might influence model performance. During the clarification phase, they collaboratively discuss and refine their understanding of key terms like hyperparameters, accuracy, and cross-validation. In the questioning phase, students pose and explore analytical questions such as why model accuracy varies with changes in parameter values. Finally, in the summarization phase, they run Grid Search using tools like `GridSearchCV` in Python, analyze the results, identify the best parameter combinations, and present conclusions on how tuning improved performance. This method promotes critical thinking, peer learning, and a deep conceptual grasp of model optimization through both discussion and practice.

- **CO – PO / PSO mapping:**

CO	PO2	PO3	PO4	PSO
CO4	2	2	3	2

(1 – Low

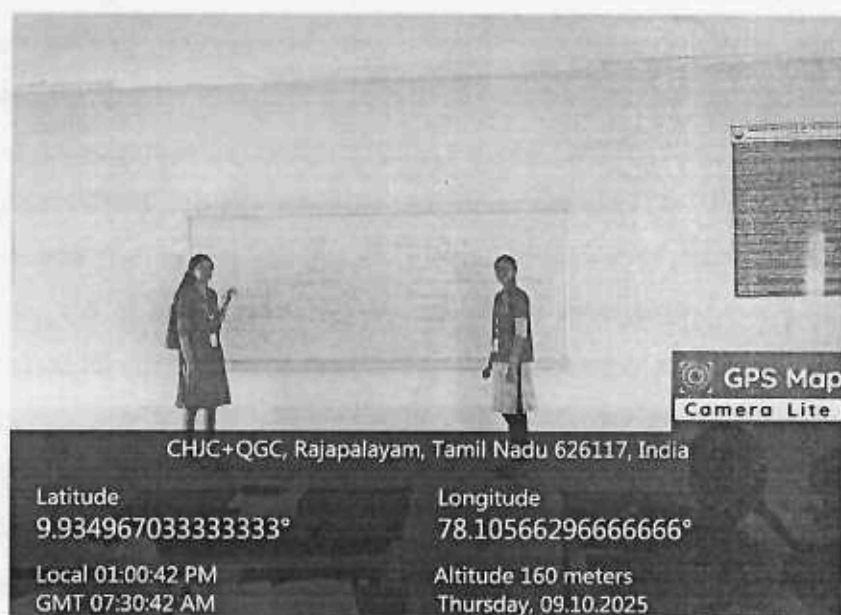
2 – Moderate

3 – High)

- **PO / PSO mapped:**

Innovative practice	
Justification for correlation	PO2 – Moderate: Students identify and apply proper evaluation strategies (such as accuracy, precision, and validation techniques) to analyze and compare deep learning models tuned using Grid Search. PO3 – Moderate: Students design and implement effective solutions by selecting and tuning hyperparameters through Grid Search to enhance model performance and efficiency. PO4 – Strong: Students conduct systematic investigations using Grid Search to study the impact of parameter variations on model outcomes, demonstrating analytical and experimental reasoning. PSO2 – Moderate: Students evaluate how tuning parameters through Grid Search improves the accuracy and performance of IT-oriented deep learning models, linking theoretical understanding to practical applications.

- **Images / Screenshot of the practice:**



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

https://docs.google.com/forms/d/13y4j7_heSVzD4m0dRPc36uw3YnqoQrSOGCZKvtjwzJ0/edit

Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

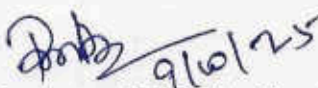
- ❖ **Improves model accuracy and performance** by identifying the best hyperparameter combinations.
- ❖ **Develops analytical and experimental skills** through systematic investigation of model behavior.
- ❖ **Bridges theory with practical application**, helping students understand how tuning impacts real-world deep learning results.

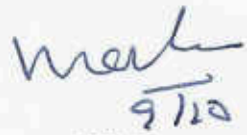
Challenges faced in implementation:

1. Evaluating all parameter combinations can be time-consuming and resource-intensive, especially with large datasets or complex models.
2. Without proper validation, Grid Search may select parameters that perform well on training data but fail to generalize to new data.
3. Choosing appropriate parameter values and ranges requires prior knowledge and experimentation, which can be difficult for beginners.

References:

1. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html?utm_source=chatgpt.com
2. <https://www.youtube.com/watch?v=lfw2Rh2v8k&t=86s>


Signature of Faculty Member


HOD



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Department of Artificial Intelligence and Data Science

Academic Year (2025-2026) ODD Semester

Innovative Teaching Practice

Course Code/Name : AD3501- Deep Learning

Year/Sem/Sec: III/V/B

Teaching Methodology :Reciprocal teaching

Date:09/10/2025

Faculty Name : Dr R M Rajeshwari

Topic : Grid Search in Deep Learning

Objective:

To understand the concept and working of **Grid Search** — a hyperparameter tuning technique — through **peer discussion and reciprocal teaching**.

Teaching Method:

Reciprocal Teaching – a cooperative learning strategy where two students take turns being the **teacher** and **learner**.

They explain, question, summarize, and clarify the topic together.

Participants:

- **M Jameel Fathima (Teacher 1):** Explains the concept of Grid Search.
- **S Aswitha (Teacher 2):** Asks questions, summarizes, and clarifies understanding.

Teaching Dialogue / Activity Script

M Jameel Fathima:

"Let's start! Grid Search is a method used in machine learning to find the best hyperparameters for a model.

For example, when we use an SVM model, we have to choose parameters like *c* and *kernel*. Grid Search tests all combinations of these to find the best one."

S Aswitha:

"So it tries all possibilities? That sounds like testing every setting?"

M Jameel Fathima:

"Exactly! That's why it's called *Grid* Search — it creates a grid of all parameter values. For instance:

$C = [0.1, 1, 10]$ and $kernel = ['linear', 'rbf']$

→ It tests $3 \times 2 = 6$ combinations."

S Aswitha:

"Got it! But how does it know which one is best?"

M Jameel Fathima:

"It uses **cross-validation** — each combination is trained and tested on data, and the one with the **highest accuracy** is selected."

S Aswitha (summarizing):

"So Grid Search = testing all parameter combinations + cross-validation → best score = best parameters."

M Jameel Fathima:

"Perfect summary! But remember, it can be slow when there are too many parameters — that's why sometimes we use **Random Search** as a faster alternative."

S Aswitha:

"Thanks! I'll explain it back to you now."

S Aswitha (teaching back):

"Grid Search helps tune parameters by systematically trying all combinations. It checks model performance for each using cross-validation, and finally picks the best settings for highest accuracy."

M Jameel Fathima:

"Excellent — you've got it!"

Learning Outcome:

Students understood:

- What **Grid Search** is.
- How it works (parameter grid + cross-validation).
- Its **advantages** (systematic and accurate).
- Its **limitations** (time-consuming for large grids).

Conclusion:

Through **reciprocal teaching**, both learners actively participated, questioned, and clarified the working of **Grid Search**.

This peer-learning method improved conceptual understanding and communication skills while reinforcing the idea of **hyperparameter tuning**.

1. Did both students clearly explain what *Grid Search* is and how it helps in selecting the best hyperparameters?

☐ Yes

☐ No

2. Was the step-by-step process of Grid Search (forming the grid, testing combinations, and choosing the best) easy to follow?

☐ Easy

☐ Moderate

☐ Difficult

3. Did the students use clear examples to make the concept understandable?

☐ Yes

☐ No

4. Did both students participate equally and ask meaningful questions to check each other's understanding?

☐ Yes

☐ No

5. After watching the reciprocal teaching, how confident are you in explaining Grid Search on your own? (Rate on a scale of 1–5: 1 = not clear, 5 = very clear)

a. ☆ b. ☆ ☆ c. ☆ ☆ ☆ d. ☆ ☆ ☆ ☆ e. ☆ ☆ ☆ ☆ ☆


Signature of Faculty

Dr R M Rajeshwari

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9/10

HOD

Dr M Kaliappan